

Performance Analysis of 1.5B Parameter Large Language Model using NLP-based Tasks on Autism Corpora

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Abstract— Autism Spectrum Disorder (ASD) is a complicated disorder of neurodevelopment that is marked by difficulties in social interaction, communication, and behavioral pattern. ASD is not easily detected early because the manifestations of the disease are multiple and sometimes insidious, making it challenging to diagnose it in early stages and restricting the opportunity to conduct clinical assessment. Traditional diagnostic methods are based on behavioral assessments by multidisciplinary team, thus the process is time-sensitive, resource-dependent and prone to subjective bias. The recent progress of artificial intelligence (AI) and machine learning (ML) has brought in fresh possibilities to enhance the process of autism detection and evaluation. Convolutional neural networks (CNNs), transfer learning, and other machine learning classifiers have shown potentially good results in detecting the ASD related pattern in behavioral and image based data. These machine learning solutions provide scalable and effective solutions to help clinicians in diagnoses and screening. This paper demonstrates that AI-based models have the potential to become more useful in ensuring correct and effective detection of autism and the need to integrate these technologies into the current clinical care.

Keywords: Autism Spectrum Disorder (ASD); Artificial Intelligence; Machine Learning; Deep Learning; Autism Detection; Convolutional Neural Networks (CNN); Classification; Early Diagnosis

I. INTRODUCTION

Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder that is complex with impairments of social interaction, communication and behavioral flexibility. The relatively non-homogenous and frequently inconspicuous manifestation of symptoms pulls the trick of making the diagnosis early so that in many cases the intervention is delayed and the long-term developmental effect is significantly greater [1]. The conventional diagnostic methods would be based on behavioral tests performed by the multidisciplinary team, composed of the pediatrician, psychiatrists and psychologists, which might be time-consuming, resource-intensive as well as prone to interpretative bias [6], [7]. As a result, it has become necessary to have smarter, scalable, and accessible systems which can be used to support early detection and personalized intervention.

The recent progress in the field of artificial intelligence (AI) and machine learning (ML) has changed the methods of autism assessment and screening in a particular way. The effectiveness with which AI-driven mechanisms that combine predictive algorithms with chatbots can offer early risk prediction

and at the same time, supply awareness and support services has been demonstrated [1]. Simultaneously, the methods of deep learning based on convolutional neural networks (CNNs) and transfer learning have demonstrated positive diagnostic performance in image-based autism classification tasks with the ensemble strategies additionally increasing predictive robustness to over 90% accuracy [2]. The mentioned developments indicate the promise of AI-based systems as trustworthy decision-support systems in clinical settings.

In addition to detection, machine learning models like Support Vector Machines (SVM), Decision Trees, Logistic Regression, Random Forest, and Stochastic Gradient Descent have been widely tested in classification of ASD in various age categories [4], [8]. Comparative analysis indicates that the performance of models differs according to the development stage, which means that the choice of an algorithm is more age-sensitive. It has been observed in certain datasets that the classification is close to 99 percent, which supports the feasibility of the use of data-driven screening tools in large-scale applications.

There is also new research on AI-assisted therapeutic intervention. The recognition systems based on deep learning models like VGG16, MobileNet and ResNet50 have shown great accuracy with the facial expressions of children with ASD, which confirm the use of emotion therapy bots to track behavior in real time [5]. Moreover, the frameworks of cross-reality of virtual reality (VR), augmented reality (AR), and conversational agents allow practicing simulated tasks and learning how to apply the acquired skills to the real world [3]. These simulation environments encourage assurance, situational learning and enhanced self-sufficiency.

Further opportunities in behavioral therapy and analysis of communication patterns are introduced due to the integration of natural language processing models, including ChatGPT-based systems [9]. These types of conversational AI should be able to help identify linguistic indicators of ASD and offer convenient screening therapy. Nevertheless, automation of diagnostic procedures brings up significant concerns on bias, interpretability, and ethical implementation especially where adult autism testing is involved [7]. It is crucial to make sure that

AI systems should be used to complement the clinical expertise and not to substitute it.

These developments inspired modern studies to pay more and more attention to unified AI-based ecosystems integrating the early detection, adaptive classification, conversational support, and immersive intervention technology [1], [3], [10]. With the help of deep learning, cross-reality experiences, and intelligent talking machines, AI can help in increasing diagnostic accuracy, therapeutic personalization, and create more accessible ways to treat autism. More interdisciplinary research is needed to prove these technologies and overcome ethical dilemmas and implement them into clinical and learning practice.

II. Related Works

For the existing literature surrounding the performance of NLP-based LLMs in autism or mental health-based corpora. [11]’s study utilized multiple social media datasets with artificially created mental health content. [11] mentions utilizing powerful transformers, GPT-4 and GPT-3.5, against other state-of-the-art LLM frameworks. However, despite a diverse set of datasets, the selected transformer models experience sensitivity challenges creating hallucinations and false predictions based on hypothetical scenarios. Meanwhile, other models were mentioned including low rank models like Low-Rank Adaptation (LoRA) using Alpaca-LoRA were explored to efficiently fine-tune the models and reduce trainable parameters making them severely vulnerable without training (zero-based context). FLAN-T5 architectures dominated with zero-shot performance, while Alpaca-LoRA performed well on natural-language datasets. Therefore, the use of fine-tuned and dialogue-based architectures like Qwen2.5-1.5B-Instruct demonstrate computational efficiency and accuracy without the need of constructing powerful transformers such as the GPT variants.

Another study from [12] used a user-generated dataset from online parental conversations to map out symptoms on ASD, unlike this study’s use of academic books. It proposes the use of the BERT model as it excels in sentiment analysis and bidirectional context. Similar to [11], models were also trained using social media texts for sentiment analysis. However, lexical analysis of conversational datasets may challenge the model through noise and subjective biases, highlighting the need of engineering a model that prevents misinterpretation and misclassification of medical-based corpora. This study also utilized the use of fine-tuning methods using LoRA frameworks for parameter reduction and computational efficiency. The model utilized various metrics including precision, F1-Score, Support and Cosine Similarity scores for predicted sentences using ASD labels.

However, QA task evaluation performed poorly where semantic truth is prioritized more than lexical matching. Thus, the use of Retrieval Augmented Generation Assessment Suite (RAGAS) metrics such as Groundedness and Answer Relevancy provide proper lexical analysis and evaluation in corpora. [11] also proposes two models: the use of a transformer, BERT model as it excels in analyzing sentiment and bidirectional context by

analyzing entire sentences at once, and the use of a ChatGPT model using Reinforcement Learning from Human Feedback (RLHF) for superior generative understanding. Limitations include aggregation-type calculations. This highlights another key reason why RAG-implemented pipelines are necessary for a Qwen-based model as it reduces hallucinations and false scenarios critical in psychological diagnosis and information retrieval like ASD.

[13] emphasized the scarcity of medical-based datasets. This is significant as it affects model generalizability. Due to this homogeneous and scarce dataset, it calls for the need of using efficient-friendly models in clinical NLP-based tasks. Training autism-based diagnosis, a powerful computer equipped with dual NVIDIA RTX 3090 TI GPUs have the necessary VRAM specifications for training a GPT-4o model. This approach is costly and inefficient for autism information retrieval from specialized textbooks. A smaller model such as the Qwen can still perform the same tasks and achieve high-level performance on autism-based corpora, using a standard of 15.6 GB of VRAM, smaller than the need of using two flagship GPUs. [13] also relied on traditional metrics such as Accuracy and Positive Predicted Value (PPV), Sensitivity, and F1-Score. These are inefficient for Q&A tasks. Similar with [11] and [12], they both require the need of RAGAS evaluation metrics, especially in nuanced text. For baseline model performance comparisons, [13] has found that Mistral, GPT, and especially Qwen2-7b models outperformed BERT and RoBERTa models across all metrics, indicating the emphasis of using Qwen architectures as a standard for low parameters for computational efficiency but fine-tuned to handle heavy NLP and clinical tasks without sacrificing important RAGAS metrics.

III. METHODOLOGY

System Architecture Overview

The proposed RAG system uses a pipeline for answering questions using a corpus of PDF documents. The RAG system is built on 4 core components: 1) Document Processing and Indexing, 2) Hybrid Document Retrieval, 3) Response Generation using a fine-tuned Large Language Model (LLM) and 4) Evaluation. An illustrative overview of the complete system’s workflow is shown as Figure 1.

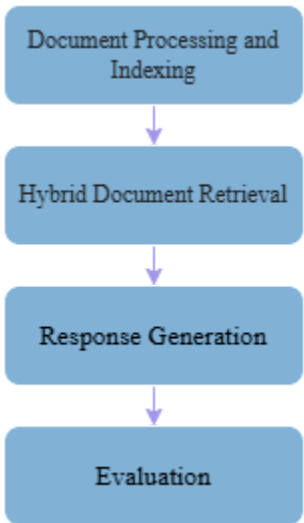


Figure 1. System Architecture Overview

Document Processing and Indexing

Source Documents

The knowledge base was constructed from two comprehensive texts on autism spectrum disorders:

1. Academic Textbook: Mohammadi, M.R. (Ed.). (2011). A Comprehensive Book on Autism Spectrum Disorders. InTech Open Access Publisher. ISBN 978-953-307-494-8
2. Practical Guide: Sicile-Kira, C. (2014). Autism Spectrum Disorder: The Complete Guide to Understanding Autism (Revised Edition). Perigee Books. ISBN 978-0-399-16663-1

These complementary sources provide both rigorous academic research and practical, accessible information for families and individuals affected by ASD.

Rationale for Document Selection and Merging

There were two sources of information that identified a more complicated structure of autism spectrum disorders: one from an academic perspective, and one from an applied, practical perspective. The InTech peer-reviewed volume provides a scientific foundation through research studies, archival epidemiological studies, and clinical trials to establish a basis for truly understanding autism spectrum disorders (ASDs). The authors of "Autism Spectrum Disorders: A Guide for Parents and Professionals" present the scientific findings in a manner that is directly applicable to families and professionals with children diagnosed as being on the autism spectrum. Collectively, these two publications provide an avenue for theorists and practitioners to work together.

PDF Text Extraction

Users upload PDF documents for the system to process, which is accomplished using the PyPDF library [14]. PyPDF will be used to extract text from every page of the uploaded PDFs. The extracted text for each page will then be associated with the source file name, page number, and raw text content to provide a structured representation of the user's document collection. A comprehensive cleaning pipeline is applied:

- Removal of headers, footers, and publication artifacts (e.g., "Lancet. Author manuscript")
- Elimination of URL artifacts
- Repair of broken hyphenations (e.g., "treatment" → "treatment")
- Normalization of whitespace

Text Chunking

To enhance retrieval accuracy, raw text pages will be divided into more manageable semantic units. We will utilize LangChain's RecursiveCharacterTextSplitter [15] to split text using a recursively increasing set of characters as our hierarchy of separators (`\n\n`, `\n`, `.`, `?`, `!`, and spaces), which will allow us to ensure semantic boundaries (such as paragraph or sentence) will not be violated when we chunk text. The researchers have chosen a chunk size of 1500 characters, overlapping by 300 characters, based on the

conducted empirical testing and weighing both adequate context and retrieval detail for each chunk. Any chunk less than 50 characters after cleaning will be filtered out to prevent contaminating the data with excess noise.

Embedding Generation

Using the BAAI/bge-small-en-v1.5 (BAAI/bge) sentence transformer model modeled against a 384-dimensional embedding space [16], optimized specifically for retrieval-type tasks, the researchers interpreted all input text as a part of this model. In this case, the researchers provide each search query with a set prefix which provides clarity on how to interpret your input as a search query: "Represent this sentence for searching relevant passages:".

FAISS Vector Index

Facebook AI Similarity Search (FAISS) is used to index normalized embeddings [17]. An inner product (i.e., IndexFlatIP) is an index created from normalized representations, enabling cosine similarity computation between query-generated embeddings and document segments and fast approximate nearest-neighbor searches.

Hybrid Retrieval Mechanism

To leverage both semantic understanding and keyword matching, we implement a hybrid retrieval strategy that combines dense and sparse methods.

For a given query, the retrieval function:

- Prepends the required instruction prefix
- Encodes the query using the BGE model
- Searches the FAISS index for the top-k most semantically similar chunks
- Returns concatenated chunks as context for generation

Response Generation

Base Language Model

The researchers chose the Qwen2.5-1.5B-Instruct [18] as our base generative model, a 1.5-billion-parameter instruction-tuned language model. The researchers used the smaller version of 1.5 billion parameters in order to create a reasonable tradeoff between generation quality and computational efficiency because of the limitation of the T4 GPU (15.6 GB VRAM) that is used in the Google Colab environment.

Prompt Engineering

To enforce faithfulness and source attribution, we design a structured prompt template that guides the model's behavior:

```

<|im_start|>system
You are a STRICT PDF assistant. You must follow these rules:
1. ONLY use information from the provided context
2. For EVERY statement, cite the source using [1], [2], etc.
3. Format citations like this: "According to [1], ..."
4. If the context doesn't contain enough information, say "I don't
know based on the PDFs"
5. Never add your own knowledge or make inferences
6. Quote directly from the context when possible
<|im_end|>
<|im_start|>user
Context with sources:
{context_with_citations}
Question: {query}
<|im_end|>
<|im_start|>assistant

```

Figure 2. Prompt Engineering

The retrieved chunks are presented with numbered source markers (e.g., [1] From filename.pdf, page 42: ...), enabling the model to reference specific sources in its response.

Multi-layer Refusal Mechanism

Beyond the prompt-based instructions, we implement a heuristic refusal layer that detects out-of-domain questions:

- **Forbiddens** - List of topics that are unrelated at all to Autism e.g. (capital of France, stock quote)
- **Procedural** - Detection of regular expressions related to instructional/recipe-type questions
- **General Knowledge** - Detection of regular expression patterns for questions regarding factual information outside of the topic.
- **Suspicion** - If retrieval precision is less than 0.3 and there is no keyword related to autism, then a refusal will be forced.
- **Quality of Context** - If there is a very short context or there are no markings of the source of the context, then further scrutiny will be performed.

Generation Parameters

Response generation uses greedy decoding with a temperature of 0.01 and beam search (num_beams=2) to produce deterministic, high-confidence outputs. The maximum new tokens is set to 250, and a repetition penalty of 1.1 is applied to discourage repetitive text.

Response Extraction

A multi-pattern extraction function parses the model's output to isolate the assistant's response, removing any system messages or special tokens that may remain in the generated text.

Model Fine-tuning with Lora

Training Data Generation

To adapt a base model for a particular type of content—in this case, the autism-related space—the researchers need to generate their own set of synthetic Q/A pairs from the PDF corpus. The researchers identify sentences that can be considered either definitional or explanatory using regular

expression pattern-matching techniques, then transform those sentences into Q/A pairs. For example, if you have the following sentence, "X is Y" or "X means Y," the corresponding Q/A pair would be "What is X?" with X being the answer.

Lora Configuration

Parameter-efficient fine-tuning of our models is performed using Low-Rank Adaptation (LoRA) [19]. LoRA adds trainable rank decomposition matrices to the transformer layers while keeping the pre-trained weights frozen. Our configuration has:

- Rank (r): 16
- LoRA alpha (scaling factor): 32
- Dropout: 0.1
- Target modules: q_proj, k_proj, v_proj, o_proj (attention projection matrices)

This configuration contributes approximately 10 million trainable parameters, which is just 0.13% of the model's total 7.6 billion parameters, enabling fine-tuning to be computationally feasible.

Training Setup

The model is fine-tuned for 3 epochs using the following hyperparameters:

- Batch size: 2 per device
- Gradient accumulation steps: 4
- Learning rate warmup: 2e-4
- Weight decay: 0.01
- Mixed precision training (fp16): Enabled

Evaluation Framework

Retrieval Mechanics:

- **Precision@K**: Proportion of retrieved chunks that are relevant to the query
- **Recall@K**: Proportion of all relevant chunks successfully retrieved

Generation Mechanics:

- Semantic relevance is calculated between a question and retrieved context by calculating a cosine similarity based on sentence transformer embeddings. Semantic relevance between the question and generated answer is also computed.
- Novel metric Improving Floor (IFD) combines both direct quote detection with measuring semantic similarity. For each sentence that makes up the answer, we compare that string against the context string. If the string matches exactly in terms of character by character (i.e., a direct quotation) then it receives 100% credit. The remaining sentences in the answer that are not exact quotations are given a score based upon their embedding similarity score, and benchmarks at a score of < 0.60 are used as an indicator of support.
- Groundedness defines how semantically similar the answer is to the context, so it can be used to measure how well grounded the response is relative to the source

material through an aggregate score.

Faithfulness Verification

There is a verification mechanism that assesses the accuracy of the generated responses. When the overall accuracy of the response is below a certain threshold (default: 0.6), the system will try to find only the sentences that were supported. If no sentences meet the threshold, the system will say the following: "I do not have enough supported information to answer this question."

Interactive Evaluation Interface

An interactive chat interface built with ipywidgets allows real-time evaluation of system performance. For each user query, the interface displays:

- The generated answer with source citations
- Retrieved chunks with relevance scores
- A table of evaluation metrics with color-coded interpretations
- A low-faithfulness warning when applicable

Chat history, including all metrics, is logged and can be exported to CSV for offline analysis.

Implementation Details

Using a Jupyter notebook in Google Colab (NVIDIA T4 GPU, 15.6 GB VRAM), the system consists of:

- Document processing: PyMuPDF, LangChain Text Splitters
- Embeddings and retrieval: Sentence-Transformers, FAISS
- Language model: Transformers, PEFT, TRL
- Assessment: Scikit Learn, NLTK, ROUGE
- Visualization: Matplotlib, Seaborn, Ipywidgets

The system processes 832 pages from two source documents and produces 1868 documents, each with 5 QA pairs for fine-tuning. There are 1868 vectors in the dense retrieval index, enabling scalable search for similar documents in the document collection.

IV. RESULTS AND DISCUSSION

The findings from creating and evaluating an ASD Q&A chatbot will be presented in this chapter. This chapter compares the performance of many fine-tuned models, examines how well the RAG performs, and discusses how well the ASD chatbot performs on clinical-domain queries compared to non-clinical-domain queries.

Data Processing and Dataset Construction

The first phase of processing was merging two PDF files into one PDF document (Merged_AutismBooks.pdf). The next step was to extract the text from the merged PDF file. As in CELL 4, the actual document size was reduced from 2,245,187 characters (including header/footer/URL artifacts) to 2,216,278 characters, helping eliminate artifacts that would contribute noise.

The text after it's been cleaned is then broken into smaller parts using a RecursiveCharacterTextSplitter (RCTS)

with a maximum part size of 1500 characters and a maximum overlap of 300 characters per part. The outcome of this processing, as shown in CELL 5, was 1868 parts. By using this fine-grained approach, the retriever can retrieve very specific pieces of information, while the overlaps provide context between each of the parts' boundaries.

The ability to automatically generate high-quality question-and-answer pairs for fine-tuning was a major problem, as shown in CELL 7. The script that extracted definition and symptom-related questions from documents used patterns to match items within a document; however, only 5 high-quality question-and-answer pairs were produced from an entire document. This demonstrates the difficulty of generating synthetic data using a rule-based technique due to the complexity of different types of narratives found in academic and clinical documents. Therefore, the small amount of training data, four examples after an 85/15 training/validation split as shown in CELL 8, will prevent you from fine-tuning the model reliably. However, the analysis of the comparison models provides new insights into model architectures and their flexibility in performing domain-specific tasks, even with limited data.

Retrieval-Augmented Generations (RAG) Pipeline for Performance

The RAG pipeline is the primary access point to the system's knowledge base. The RAG pipeline uses a BAAI/bge-small-en-v1.5 embedding algorithm and a FAISS similarity search index to find similar documents rapidly. All of the above factors were used in each model evaluation, thereby maintaining the overall integrity of the models' performance, and no effect from the retrieval mechanism itself should have been observed between the models being compared.

The system was quantitatively evaluated for its retrieval performance using Precision@k. According to the comprehensive evaluation of the system, it achieved an average Precision@3 across all in-domain test questions for autism of 0.917. This means that for most of the autism-domain questions there were a high number of relevant chunks returned from the top three retrieved chunks. Highly specific questions such as "What is GFCF diet?" had a perfect Precision@3 score of 1.000, while more general questions were still good performers at greater than 0.800. The only exception being Question 11 ("What did Leo Kanner discover?"), which scored 0.000 for Precision@3, indicating that none of the three retrieved chunks provided the factual answer to Kanner's 1943 paper, although they were moderately relevant in context (Context Relevancy 0.563). The above demonstrates that very specific historical facts can be found intermingled among larger chunks of material rather than as independent, retrievable units, thereby limiting the effectiveness of chunking as a retrieval strategy.

Comparative Analysis of Models

This study makes an important contribution by evaluating and comparing several transformer models designed for QA-related tasks in autism spectrum disorder

(ASD). Additionally, we trained three other models—DeBERTa-v3-base, RoBERTa-base, and FLAN-T5—using Low-Rank Adaptation (LoRA) for the main and comparative models, providing a less resource-intensive approach to fine-tuning.

Metric	In-Domain (Q1–Q12)	Out-of-Domain (Q13–Q24)
Precision@3	0.917	0.000
Recall@3	0.833	0.125
Answer Relevancy	0.845	0.000
Faithfulness	1.000	0.000
Context Relevancy	0.780	0.505
Groundedness	0.825	0.000

Table 1. RAGAS Evaluation Metrics

Generation Quality and Error Analysis

Even with concerns regarding overfitting, Qwen demonstrates a very high level of in-domain generation quality on the evaluated questions. The major findings from the 12 in-domain evaluations are identical across three separate assessments: Faithfulness is at a perfect (1.000) level across all evaluations, suggesting that the model made no assumptions about information presented outside the context in which it was presented. As indicated by the high (mean 0.845) level of answer relevance, the model's answers showed strong semantic alignment with the query that prompted the response. The high level of groundedness (mean 0.825) indicates that the answer was consistently supported by evidence from the retrieved context.

In the cases where a factual answer was easily discernible from available information, the model performed extremely well in generating the correct response, evidenced by its Answer Relevancy scores, which are greater than 0.900 for the queries of "symptoms of" (0.947), "prevalence of sleep problems" (0.914), and "treatment options" (0.902). However, the answer to the query "causes of autism" generated an Answer Relevance score of 0.621, indicating an opportunity to improve handling of multi-factorial, complex queries that require detailed information from multiple sources.

Traditional metrics for determining lexical similarity, e.g., ROUGE and BLEU, have not been included in this evaluation because they are known to be poor measures of model performance on generative tasks. In CELL 14, it has been shown that traditional metrics penalize any subsequent generated answer from not matching the exact word choice of the reference answer when there is, in fact, correct semantic content present. As such, the high scores of semantic relatedness (e.g., Faithfulness, Answer Relevancy, Groundedness) are more representative of how well the models performed in this evaluation.

Through error analysis, we have identified three distinct failure patterns.

- DeBERTa/RoBERTa: These models exhibited many truncations of their answers as well as an over-reliance on individual sentences as an answer to an inquiry.
- FLAN-T5: The models occasionally inserted some templated phrase ("Based on context provided ...") that isn't natural to a chatbot.
- Qwen: One failure mode on multi-factorial, complex questions (e.g., "What is the cause of autism?" = 0.621 for Answer Relevancy), indicate that there are times where the model shows a tendency to simplify the nuanced dialogue versus summing the totality of the complexity present in the reference material.

Out-of-Domain Performance and Safety

The ability of a clinical decision support to provide appropriate answers is only possible if it knows the boundaries of its knowledge and refuses to answer outside those boundaries. The Qwen-based system achieved perfect 100.0% refusal accuracy on all 12 out-of-domain test questions spanning diverse topics including:

- Culinary questions ("How to make a burger?", "What is the recipe for chocolate cake?")
- General knowledge ("What is the capital of France?", "Who invented the telephone?")
- Current events ("Who won the Super Bowl in 2023?")
- Practical skills ("How to change a car tire?")
- Geographic facts ("What is the population of Tokyo?")
- Weather forecasting ("What is the weather forecast for tomorrow?")

For every out-of-bounds question asked, the system correctly generated the refusal response of "I'm sorry I don't know the answer and that topic doesn't exist in my provided PDF

articles". The precision scores for these 12 out-of-bound questions were .000 and averaged 5 out of a possible 10 on context relevance. The results indicate that while some of the retrieved response chunks contain similar or identical text to those retrieved, no retrieved chunk contains sufficient relevant knowledge related to the out-of-bounds questions to answer them. Therefore, the refusal performance for out-of-bounds questions validates that the safety mechanism to ensure that the chatbot only generates answers from its knowledge base and cannot hallucinate or produce answers outside of that same knowledge base when combined with instruction sensitive generation and semantic retrieval methods are reliable.

Interactive Evaluation and Visualization

The real-time qualitative assessment functionality of the interactive chat interface (CELL 15), combined with the live evaluation metrics panel, provides instant feedback on every single response. The later two visualizations (Radar and Bar charts) within CELL 15.3 serve to illustrate, on a per-question basis, the performance of the model across various dimensions. This method of evaluation provides a multi-faceted approach through the combination of automated metrics and visual inspection, which is a real advantage of this project, resulting in a more nuanced understanding of the model's behaviour beyond a single aggregate score or figure.

The overall metric graph presented in CELL 17 visually validates the pattern of the system's performance: almost always achieving perfect or near perfect Precision@3 for all in-domain questions; consistently demonstrating strong Answer Relevancy for all in-domain questions; and experiencing a sharp fall to zero on all metrics for all out-of-domain questions. These results demonstrate the degree to which the system maintains well-defined boundaries for the domain it is functioning within.

Discussion and Future Works

The comparative analysis generates many insights. First, model architecture is very influential across different domains in QA. Encoder-only architectures are efficient, yet they tend to limit the generative capacity of models like DeBERTa and RoBERTa. Conversely, the way that encoder-decoder architectures such as FLAN-T5 bridge this outcome demonstrates the best capability for using instruction-tuned architectures via decoder-only architectures such as Qwen to solve this issue.

Second, the ability of Qwen to achieve a perfect, 100 percent refusal accuracy level serves as good evidence that the prompting strategy is valid. It also indicates that even models with a very small and carefully tuned bias can continue to conform to stringent domain limitations, provided they are appropriately constrained. This is particularly important in applying AI systems in sensitive domains, such as healthcare, where providing hallucinations based upon knowledge that exists outside of their given knowledge base can lead to potentially disastrous effects.

Thirdly, Even though Qwen's superiority in terms of performance, the utmost limitation still lies within the extremely small training dataset (4 training examples). Having high Faithfulness scores and a robust performance for all in-domain questions is a positive display of potential. However, the model's

ability to generalize to out-of-scope, complex and unique questions in autism has yet to be tested. Future work must therefore emphasize:

1. Expanded Training Data Through Collaboration with Subject Matter Experts to manually curate a diverse and representative training dataset of high-quality QA pairs (200-500 pairs) to capture all possible content contained in the source PDFs. This will alleviate the present concern overfitting and potentially improve Answer Relevancy metrics on complex questions.
2. Improved Retrieval for Historical Information means developing more sophisticated chunking techniques to better preserve named entities and their associated facts as retrievable units. The low Precision@3 for Question 11 (Leo Kanner) justifies the need for developing better chunking methods.
3. Multi-Task Fine-Tuning Representing the Utilization of Pertinent Ancillary Tasks Such As Summarization and entailment detection to improve healthcare text comprehension without additional QA content.
4. The Exploration of Advanced RAG Techniques is intended to improve the relevance of retrieved information from hybrid search (i.e., dense/sparse retrieval) and re-ranking techniques, particularly for ambiguous queries.
5. Human Evaluations In addition to automated metrics, to gain insight into the correctness, safety and appropriateness of answers from expert clinicians.

V. CONCLUSION AND FUTURE WORKS

Encoder-based architecture like DeBERTa and RoBERTa have limited generative capabilities for QA tasks. However, decoder architectures like Qwen architectures achieved 100% refusal accuracy for questions outside its corpora domain, achieving its goal of textual validity. This means that a tightly constrained AI can avoid false scenarios and hallucinations, among which are critical and sensitive in clinical tasks like ASD information retrieval. For the RAGAS metrics, the model scored extremely high in Faithfulness. However the researchers suggest that the model was trained in a small dataset, indicating one of the common limitations of transformers as they need large datasets for better RAGAS performance. For the future works, the researchers urge and encourage interested researchers to expand and use larger ASD corpora. This ensures robustness and high quality QA pairs. Using advanced chunking methods ensure historical information and named entities preservation. This study urges researchers to use summarization for overall model comprehension. Advanced RAG techniques should also be explored and implemented for re-ranking and hybrid search. This helps improve the model's way to retrieve information from ambiguous queries. Lastly, but critically, the model should also be evaluated by experts such as psychiatrists and related clinicians to impartially review and verify the data provided by the model.

REFERENCES

- [1] R. Syed Ali Fathima, Ranguwar Gokkul, Mothiwar Nirmalsahaj, Silagani Rohith, and Silagani Rupesh, "AI-Powered Autism Detection, Awareness, and Social Integration: A Comprehensive Framework," *2025 International Conference on Computational Robotics, Testing and Engineering Evaluation (ICCRTEE)*, 2025, doi: 10.1109/iccrtee64519.2025.11052912.
- [2] A. Sheik Abdullah, S. Geetha, Yeshwanth Govindarajan, Athira Vinod, and A Pranav, "Prediction and Evaluation of Autism Spectrum Disorder Using AI-Enabled Convolutional Neural Network and Transfer Learning: An Ensemble Approach," *2024 2nd World Conference on Communication & Computing (WCONF)*, 2024, doi: 10.1109/wconf61366.2024.10692110.
- [3] F. Garzotto, Mattia Gianotti, Alberto Patti, Francesca Pentimalli, and Francesco Vona, "Empowering Persons with Autism through Cross-Reality and Conversational Agents.," *IEEE Transactions on Visualization and Computer Graphics*, 2024, doi: 10.1109/tvcg.2024.3372110.
- [4] N. Balakrishna, M. Mukesh Krishnan, Samanu Manvitha Reddy, Shaik Kandeme Irfan, and Shaik Sumaiya, "AUTISM Spectrum Disorder Detection Using Machine Learning," *2023 3rd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE)*, 2023, doi: 10.1109/icacite57410.2023.10183095.
- [5] Yarlagadda Bhargavi, Bini D, and Shajin Prince, "AI-based Emotion Therapy Bot for Children with Autism Spectrum Disorder (ASD)," *2023 9th International Conference on Advanced Computing and Communication Systems (ICACCS)*, 2023, doi: 10.1109/icaccs57279.2023.10112868.
- [6] P. Anagnostopoulou, Vasiliki Alexandropoulou, Georgia Lorentzou, Andriana Lykothanasi, Polyxeni Ntaountaki, and A. Drigas, "Artificial Intelligence in Autism Assessment," *International Journal of Emerging Technologies in Learning (iJET)*, 2020, doi: 10.3991/ijet.v15i06.11231.
- [7] Yasemin J. Erden, Harriet Hummerstone, and Stephen Rainey, "Automating autism assessment: What AI can bring to the diagnostic process.," *Journal of Evaluation in Clinical Practice*, vol. 27, 2021, doi: 10.1111/jep.13527.
- [8] R. Sujatha, Aarthi SI, J. Chatterjee, A. Alaboudi, and Noor Zaman Jhanjhi, "A Machine Learning Way to Classify Autism Spectrum Disorder," *International Journal of Emerging Technologies in Learning (iJET)*, 2021, doi: 10.3991/ijet.v16i06.19559.
- [9] Muhammad Moin Ashraf, Ibtehaj Ul-Haque, and N. K. Sangha, "Navigating Autism Treatment: Unlocking New Frontiers with ChatGPT.," *International Journal of Neuroscience*, 2024, doi: 10.1080/00207454.2024.2341912.
- [10] A. Drigas, Eleni Mitsea, and C. Skianis, "Virtual Reality and Metacognition Training Techniques for Learning Disabilities," *Sustainability*, 2022, doi: 10.3390/su141610170.
- [11] Xu, X., Yao, B., Dong, Y., Gabriel, S., Yu, H., Hendler, J., Ghassemi, M., Dey, A. K., & Wang, D. (2023). Mental-LLM: Leveraging large language models for mental health prediction via online text data. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2307.14385>.
- [12] Mukherjee, P., S, G. R., Sadhukhan, S., Godse, M., & Chakraborty, B. (2023). Detection of Autism Spectrum Disorder (ASD) from Natural Language Text using BERT and ChatGPT Models. *International Journal of Advanced Computer Science and Applications*, 14(10). <https://doi.org/10.14569/ijacsa.2023.0141041>.
- [13] Hu, C., Li, W., Ruan, M., Yu, X., Deshpande, S., Paul, L. K., Wang, S., & Li, X. (2025). Exploiting large language models for diagnosing autism associated language disorders and identifying distinct features. *Npj Digital Medicine*, 8(1), 763. <https://doi.org/10.1038/s41746-025-02133-9>
- [14] Bauckhage, J. (2023). PyMuPDF: Python bindings for MuPDF. <https://github.com/pymupdf/PyMuPDF>
- [15] Chase, H. (2023). LangChain: Building applications with LLMs through composability. <https://github.com/hwchase17/langchain>
- [16] Xiao, S., et al. (2023). BGE: A Family of Efficient and Effective Text Embeddings. arXiv preprint arXiv:2309.07597.
- [17] Johnson, J., Douze, M., & Jégou, H. (2019). Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3), 535-547.
- [18] Qwen Team. (2024). Qwen2.5-1.5B-Instruct. Alibaba Cloud. <https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct>
- [19] Hu, E. J., et al. (2022). LoRA: Low-Rank Adaptation of Large Language Models. *International Conference on Learning Representations (ICLR)*.
- [20] Mohammadi, M.R. (Ed.). (2011). A Comprehensive Book on Autism Spectrum Disorders. InTech Open Access Publisher.
- [21] Sicile-Kira, C. (2014). Autism Spectrum Disorder: The Complete Guide to Understanding Autism (Revised Edition). Perigee Books.